

EEG based Motor Imagery of Hands Classification using modified Sub-Band Common Spatial Pattern

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Abstract—Brain-computer interfaces (BCIs) leveraging motor imagery (MI) create intuitive systems that translate brain activity into direct control of computer applications. Motor imagery is a technique that provides simulated solutions during training for motor problems. For this study, Electroencephalography (EEG) data is collected from fourteen healthy subjects with 11 channels for two tasks with 40 repetitions each; the data were sourced from a publicly available database, ‘OpenVibe’. The present work proposes a modified Subband Common Spatial Pattern (mS-BCSP) with Linear Discriminant Analysis as a method to classify right and left hand motor imagery movements, which provides a significant accuracy of 91.79% compared to other methods. The mSBCSP features are extracted from six EEG sub-bands after band pass filtering to enhance the discriminatory features. The results suggest that the proposed method offers a reliable and efficient framework for MI classification, which can serve as a foundation for future development of BCI-controlled applications for motor rehabilitation and the multimedia industry.

Index Terms—Motor Imagery, Electroencephalography, OpenViBE, Machine learning models, Classification

I. INTRODUCTION

Electroencephalogram (EEG) measures the brain’s electrical activity either non-invasively, surface electrodes on the skin (scalp EEG), or invasively, with electrodes positioned directly on the cortical surface (electrocorticogram or invasive EEG) [1]. For most practical purposes, the scalp Electroencephalogram (EEG) has been explored as a potential tool to decode brain activity and drive machines or computers based on the field of application. Brain-Computer Interfaces (BCIs) are integrative systems that allow users to communicate with computers solely through their brain activity signals. One such Brain-Computer Interface (BCI) designing, testing, and implementation software platform is OpenViBE. It is an open-source tool made available by French Institute for Research in Computer Science and Automation (INRIA). BCIs based on OpenViBE have been used significantly in the fields of multimedia, medical (assistance to disabled people, robotics, real-time biofeedback, neurofeedback, real-time diagnosis),

etc. It is equipped with an acquisition server, a designer to design scenarios and example scenarios based on different EEG markers for development of real-time BCIs by programmers (integrating with Python, MATLAB, etc.) and non-programmers alike. It has been integrated with a virtual environment for a better immersion experience of virtual reality (VR) and control of robotic equipment or exoskeletons [2].

Motor Imagery (MI) involves the mental rehearsal of movement through reactivation of motor-action representations in working memory, governed by central motor control without physical execution [3]. While MI based BCI has previously been used for sports training, its applications in rehabilitation and recovery of stroke patients or patients suffering from motor limb weakness or paralysis are being explored extensively [3]. The difference in EEG of stroke and healthy subjects is being established, and various approaches for decoding Stroke MI EEG are being employed, but realisation of such an accessible MI based BCI system is plausible with the aid of the OpenViBE platform. This also forms the motivation for our work.

The challenges in MI EEG classification for healthy subjects are low Signal to noise ratio (SNR), high inter-subject variability, non-stationary nature of the recorded signals, limited MI data, etc. [4]. Features like Hilbert Transform (HT) [5], Common Spatial Pattern (CSP) [6], Sub Band CSP (SBCSP) [6], Enhanced CSP (ECSP) [8], Filter Bank Common Spatial Pattern (FBCSP) [7], Discriminative FBCSP (DFBCSP) [9], Phase Locking Value (PLV) etc. have been used with both conventional machine learning and deep learning algorithms for classification. Time-frequency domain features — including Short-time Fourier Transform (STFT), Discrete Wavelet Transform (DWT), and Wavelet Packet Decomposition (WPD) — are widely employed in deep learning classifiers as input data. Still, they require further preprocessing of EEG signals into spectrograms and are therefore time-consuming. The clas-

sification models used for binary or multiclass classification include Linear Discriminant Analysis (LDA), Convolutional Neural Network (CNN), Bidirectional LSTM (BiLSTM), and Gated Recurrent Unit (GRU) [20], Support Vector Machines (SVM), Long Short Term Memory (LSTM), random forest [19], Hybrid or fusion models, etc [4]. Their application for real-time BCI implementation might come at the cost of time lags, rendering the experience unsatisfactory. This paper uses several features with various machine learning and deep learning algorithms to classify hand movements with motor imagery based on right and left for healthy subjects. Among them, we have used modified Sub band Common Spatial Pattern (mSBCSP) with LDA for classification. This will further help in BCI-controlled Virtual reality for rehabilitation.

II. METHODOLOGY

The motor imagery based hand movements involve the following stages: Data acquisition, Data Preprocessing followed by feature extraction and classification. A block diagram of the current work is shown in Fig. 1. The overview of all the blocks is mentioned in the next subsections.

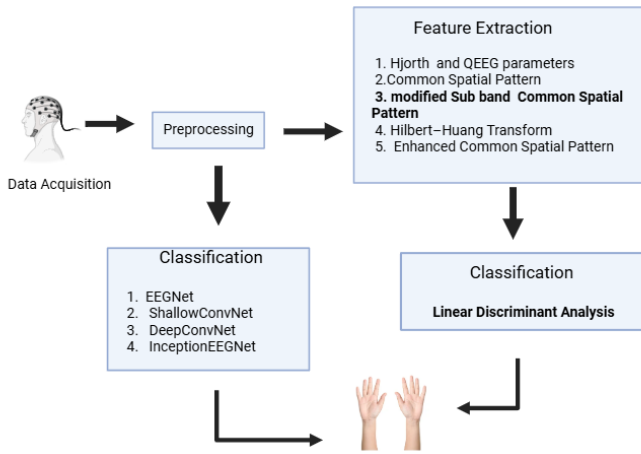


Fig. 1. Flow chart of the methodology

A. Dataset

The OpenViBE MI EEG dataset [15] has 14 recordings of right and left hand motor imagery from 11 channels, namely C1, C6, C2, Nz, FC4, FC3, C4, C3, C5, CP4 and CP3 in csv format. The channels were recorded using a Mindmedia NeXus32B amplifier in common average mode with a sampling frequency of 512 Hz, and Nz was used as a reference for further analysis. Each record had 40 trials of right and left MI task, which follows the Graz University protocol [16]. The labels were available as stimulation codes from the OpenViBE documentation.

B. Preprocessing

The data was visualized and then rearranged in Python to organize the reference channel labels and event labels into individual trials of about 6 seconds each. The channels were

re-referenced with respect to the 'Nz' channel, as shown in Fig. 2. The reference channel was then removed for further analysis. A fourth-order Butterworth bandpass filter (8–30 Hz) was applied to isolate key alpha and beta sub-bands for MI task analysis [17]. An IIR notch filter was used at 50Hz frequency to remove powerline interference with filtered output as shown in Fig. 3.

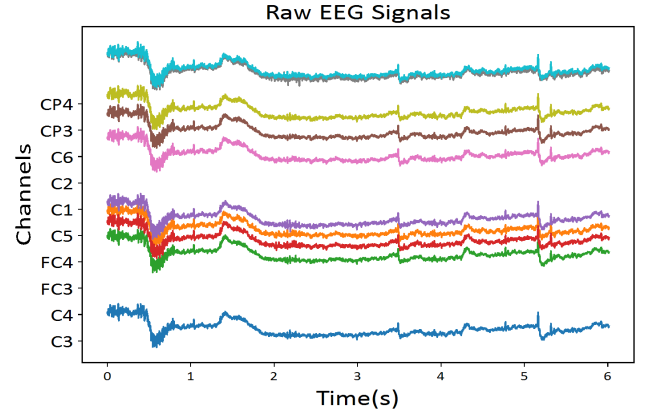


Fig. 2. Raw EEG signals of Subject 01 for trial 01

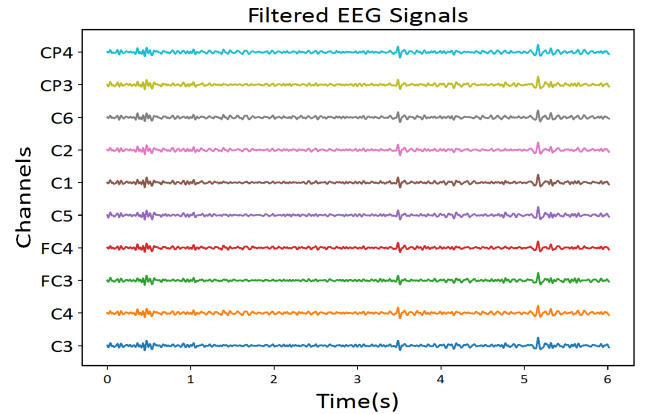


Fig. 3. Filtered EEG signals of Subject 01 for trial 01

C. Feature Extraction

After signal preprocessing, feature extraction is an essential phase in the study of EEG signals or any other biomedical signals. In this step, the dimensionality reduction and compression of data was done. This process enables the representation of data using a more compact subset of features. In our study, we have used various features: Hjorth and QEEG parameters, CSP, ECSP, HT, mSBCSP

1) *Hjorth and QEEG parameters*: For extracting these characteristics, the signals were pre-processed to capture all EEG bands from 0.5 to 45 Hz such as delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-12 Hz), beta (12-30 Hz) and gamma (30-45 Hz). These features have been previously used for various

disease states classification and monitoring of rehabilitative states [18].

- The Hjorth parameters are time domain features capturing the complex nature of signals as activity, mobility and complexity and are calculated as (1), (2), (3)

$$\text{Activity} = \text{var}(z[n]) \quad (1)$$

$$\text{Mobility} = \sqrt{\frac{\text{var}(z'[n])}{\text{var}(z[n])}} \quad (2)$$

$$\text{Complexity} = \frac{\text{Mobility}(z'[n])}{\text{Mobility}(z[n])} \quad (3)$$

where $z[n]$ is the EEG signal and $z'[n]$ represents the first-order derivative of the discrete signal.

- Quantitative EEG (QEEG) parameters serve as frequency domain features capturing power distribution along different sub-bands and comprise Absolute Power (AP), Relative Power (RP), and various ratios like Delta-Alpha Ratio (DAR), and Delta-Theta ratio (DTR), Theta-Beta ratio (TBR), and Delta-Theta-Alpha-Beta Ratio (DTABR), which were considered during this study.

$$\text{DTABR} = \frac{\text{Delta Power}}{\text{Alpha Power} + \text{Theta Power} + \text{Beta Power}} \quad (4)$$

$$\text{DAR} = \frac{\text{Delta Power}}{\text{Alpha Power}} \quad (5)$$

$$\text{DTR} = \frac{\text{Delta Power}}{\text{Theta Power}} \quad (6)$$

$$\text{TBR} = \frac{\text{Theta Power}}{\text{Beta Power}} \quad (7)$$

$$\text{AP} = \int_{f_{\min}}^{f_{\max}} P(f) df \quad (8)$$

$$\text{RP} = \frac{\text{Absolute Power of an EEG frequency band}}{\text{Total Absolute Power}} \quad (9)$$

where $f_{\max}$ and $f_{\min}$ are the highest and lowest frequencies within the specific frequency band. $P(f)$ is the power spectral density at a specific frequency.

2) *Common Spatial Pattern (CSP)*: The common spatial pattern gives new spatial components whose variances are optimal to differentiate between the two classes, right and left. The CSP features were obtained according to the methodology described in [6].

3) *Enhanced CSP (ECSP)*: The Phase Locking Value between two channels is calculated using instantaneous phase information in EEG by the Hilbert transform, which is used as a weighted matrix in order to tune the CSP features for enhancing the spatial filters. The log variance of these new modulated CSP features is used for classification [8]. It is based on the assumption that this can provide phase locking information of different brain regions concerning the channels.

4) *Hilbert Transform (HT)*: The Hilbert transform of each trial was obtained and concatenated to obtain a Hilbert spectrogram of the trial [21].

5) *modified Sub band Common Spatial Pattern (mSBCSP)*: The proposed mSBCSP algorithm used here employs a 4th order zero phase Chebyshev Type II filter to get six sub-bands 8-12Hz, 12-16Hz, 16-20Hz, 20-24Hz, 24-28Hz, 28-30Hz and extracting CSP features from each of these bands to concatenate them together without employing any feature selection steps further, unlike what has been proposed in [6] [7]. The topological maps signifying the weights of these extracted spatial patterns for classification is shown in Fig.4 for the sub bands.

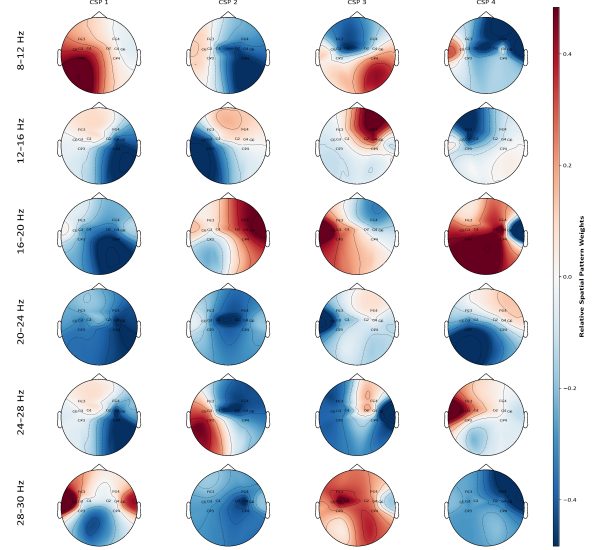


Fig. 4. Topological maps of the mSBCSP weights for classification

D. Classification

To decode the completed MI tasks the extracted features, HT, CSP, mSBCSP, ECSP, Hjorth, and QEEG parameters were individually classified by LDA. The filtered signal was classified using EEGNet [10], ShallowConvNet [12], DeepConvNet [12] and Incep-EEGNet [11]. All the results were cross-validated using five-fold cross-validation, with mean accuracy being used as a performance measure.

1) *Linear Discriminant Analysis (LDA)*: LDA is a technique that has low computation requirements that works well with linear classification tasks. It uses linear projections to increase the distance between the two class means to its maximum and reduce the variance between classes to its minimum using a separating hyperplane. Thus, it serves as an efficient classifier for MI based BCI [4].

2) *EEGNet*: EEGNet is robust, concise and generalizable CNN that has been tested across different paradigms for extracting features and classifying MI EEG tasks. The model mimics traditional EEG pipelines like Depthwise convolution for spatial filtering, Separable convolution for temporal features extraction, and a compact architecture to maintain a comparatively light-weight model. The model works efficiently

Algorithm 1 Modified Sub-band CSP (mSBCSP)

Input: $\mathbf{X} \in \mathbb{R}^{T \times C \times S}$: EEG trials with T trials, C channels, S samples
 $\mathbf{y} \in \{0, 1\}^T$: Binary class labels
 f_s : Sampling frequency in Hz
 n_c : Number of CSP components per band
 $\mathcal{B} = \{[f_{\text{low}}^{(l)}, f_{\text{high}}^{(l)}]\}_{l=1}^L$: List of L frequency bands
Output: $\mathbf{F} \in \mathbb{R}^{T \times (n_c \cdot L)}$: Concatenated CSP feature matrix

Initialize feature list $\mathcal{F} \leftarrow []$
for all $[f_{\text{low}}, f_{\text{high}}] \in \mathcal{B}$ **do**
 for $t = 1$ to T **do**
 Apply zero-phase Chebyshev Type-II bandpass filter to $\mathbf{X}_t$ in $[f_{\text{low}}, f_{\text{high}}]$
 end for
 Stack filtered trials into $\mathbf{X}_{\text{filt}} \in \mathbb{R}^{T \times C \times S}$
 Fit CSP with n_c components on $(\mathbf{X}_{\text{filt}}, \mathbf{y})$
 Project $\mathbf{X}_{\text{filt}}$ through CSP to obtain $\mathbf{F}^{(l)} \in \mathbb{R}^{T \times n_c}$
 Append $\mathbf{F}^{(l)}$ to $\mathcal{F}$
end for
Concatenate all $\mathbf{F}^{(l)}$ to form final feature matrix $\mathbf{F}$
return $\mathbf{F}$

with smaller datasets as well, which makes it suitable for OpenViBE dataset classification. [10]

3) *DeepConvNet*: The DeepConvNet [12] uses the first convolutional layers for temporal convolution and the second layer to extract spatial patterns without using activation layer between the two. These two layers perform a bandpass and spatial filtering operation, just like the conventional methods. Further, 3 convolutional blocks are added before a linear classification by the final Dense layer using softmax units.

4) *ShallowConvNet*: ShallowConvNet, unlike DeepConvNet, does the temporal convolution and the spatial filtering, after which an average pooling layer, a nonlinearity and a log activation function are introduced. These layers together are similar to the trial CSP logarithmic variance computation in FBCSP [7]. The ShallowConvNet can learn temporal features of the EEG band power changes in a single trial due to many pooling layers in a trial and a larger kernel size for the temporal convolution layer. Both these ConvNets have achieved acceptable classification accuracies compared to conventional feature extraction and classification techniques [12].

5) *Incep-EEGNet*: Incep-EEGNet is based on a CNN architecture but improves the feature extraction by using an inception stage. This concatenates features from several branches with different convolutional kernel sizes and stride lengths to eventually apply batch normalisation and activation before using softmax for classification. It has outperformed the previous feature extraction techniques used for MI classification [12]

III. RESULTS

The performance of these classification methods was evaluated for the binary classification task of left and right MI after implementing the previously explored classification techniques [4], [6]–[8], [10]–[12], [18], [21] and the proposed mSBCSP.

The 5-fold mean classification accuracies of these analysis have been listed in Table I with a maximum mean accuracy of 91.79% with the proposed method of mSBCSP + LDA classification. Fig. 5 shows the confusion matrix for the mSBCSP, LDA model that performs best, and the corresponding Receiver operating curve (ROC) with an area under the curve(AUC) score of 0.98 is shown in Fig. 6 which signifies its ability to distinguish between the left and right hand classes. The performance metrics i.e. F1- score, Precision, Recall, and Accuracy were calculated for different features with classifiers mentioned in Table I. A comparison with the previous classification results on the OpenViBE dataset is shown in Table II. This signifies a potential classification technique for MI BCI application.

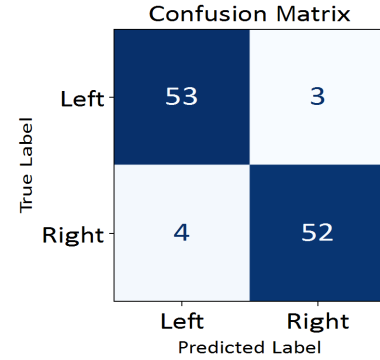


Fig. 5. Confusion matrix for the best classification using mSBCSP + LDA.

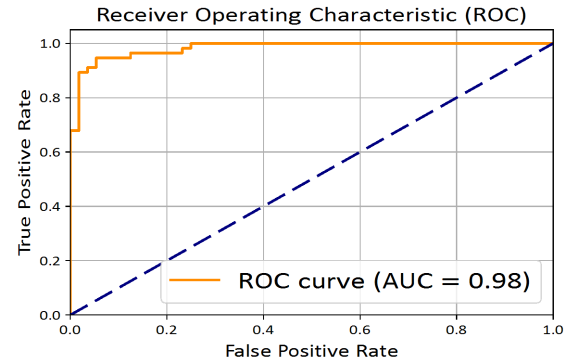


Fig. 6. ROC for the best classification achieved using mSBCSP+ LDA

IV. CONCLUSION

We analyzed the accuracy of eight existing techniques as mentioned in Table I from Sr. No. 1 to 8 for the classification of motor imagery hand movements. These techniques, though robust, had their own weaknesses to be noise prone, time consuming and there dependency on training data, which further warranted a need for exploring classification techniques. The proposed method mSBCSP+LDA shows promising results with the accuracy of 91.79% for classification of right and left hand motor imagery movements with minimal pre-processing,

TABLE I
COMPARISON OF FEATURE-CLASSIFIERS AND THEIR 5-FOLD MEAN ACCURACY

Sr. No.	Feature + Classifier	5-Fold Mean Accuracy (%)	Precision(%)	Recall(%)	F1-score(%)
1	HJT+ QEEG +LDA	71.07	71.36	71.07	70.97
2	CSP + LDA	88.57	88.66	88.57	88.57
3	ECSP + LDA	88.39	88.45	88.39	88.39
4	HT + LDA	63.30	63.18	63.04	62.94
5	EEGNet	89.46	90.08	89.46	89.41
6	DeepConvNet	89.82	90.75	89.92	89.74
7	ShallowConvNet	74.46	75.12	74.46	74.21
8	Incep-EEGNet	81.79	86.64	81.79	80.85
9	mSBCSP + LDA	91.79	91.88	91.79	91.78

TABLE II
COMPARING WITH OTHER STATE OF ART METHODS

Reference	Features+Classifier	Accuracy
Dongare et al. [13]	Auto regressive (AR) coefficients + HT+SVM	88.5%
Chakladar et al. [14]	Find Target Algorithm	70.36%
Proposed method	mSBCSP+LDA	91.79%

which can provide lower inference time for seamless BCI applications. This analysis also signifies the suitable frequency bands which could be more significant for the classification task within the broader alpha and beta EEG bands. In future, we can include more parameters and test the proposed model on larger datasets with multiple classes that would also give insights into the specific frequency bands for upper limb motor imagery. The integration of localized and global brain activity information can improve the accuracy and applicability of the brain-computer interface (BCI) based on motor imagery.

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